Lecture Replay Assistant

A course recording and review system

Searchable transcripts and source-aligned lecture replay.

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Workplace AI | Project proposal | 14 October 2026

Problem and intended users

Students reviewing a lecture

A keyword can lead to a brief mention, a recap or a preview. Students need the point where the explanation actually begins.

Teachers preparing content

Teachers review captions, choose useful frames and define lesson segments. These tasks require judgment and source checks.

Primary outcome

Find a complete explanation and inspect its source captions and board context.

Proposed first release

Review after recording

Import video and captions. Generate candidate concepts. Review the evidence. Export confirmed data for replay.

Core scope

Synchronized captions, editable concept intervals, frame snapshots and reliable export and reload.

Research extensions

Board-assisted terminology suggestions and teacher tracking. Live recording and physical camera control are outside the first release.

System architecture

Stage	Input	Output
Local preprocessing	Video and optional captions	Audio, timing and frames
Candidate generation	Transcript and available frames	Concept intervals and proposed edits
Human review	Source evidence and candidates	Confirmed captions and index
Browser replay	Video and saved index	Search and synchronized playback

Planned integration uses replaceable model adapters and a common video time base.

Application capabilities

Component	Implementation
Video playback	Browser player with playback and timeline controls
Caption workspace	Timestamped SRT text and synchronized current captions
Transcript retrieval	English and Chinese keyword search
Source navigation	Search results linked to the video timeline
Local inputs	Video / SRT selection with input validation

The application operates on prepared captions. Model-assisted processing is part of the development roadmap.

Demonstration and verification

Demonstration material

A 60-second illustrated lesson with 12 English / Chinese caption segments. Captions are manually authored and aligned to the slides.

Quality checks

Five parser and search tests passed. Browser checks covered playback, caption synchronization, timestamp navigation, bilingual queries and empty results.

Reliability

Invalid SRT input retains the existing captions. HTTP byte-range support enables reliable navigation through the video timeline.

The demonstration lesson has no audio. Model quality and learning outcomes will be evaluated separately.

Transcription and caption review

Planned transcription pipeline

Integrate speech recognition through a replaceable model interface, with timing aligned to the original video.

Source-preserving review

Retain original recognition output alongside reviewed captions so corrections can be checked against the lesson.

Evaluation approach

Use human reference transcripts to evaluate recognition errors, correction accuracy and review effort.

The current replay application accepts prepared SRT captions. Automated transcription is a planned integration stage.

Concept indexing and source evidence

Planned semantic retrieval

Associate concepts with source captions and video intervals to support searches beyond literal keyword matching.

Explanation boundaries

Distinguish a brief mention from a complete explanation using human-labelled intervals and source evidence.

Human review

Provide reviewable concept names and boundaries, with provenance retained for each indexed segment.

Concept indexing and review are development goals; the implemented retrieval workflow searches caption text.

Target hardware and deployment plan

Proposed platform

Workplace AI theme, Thunderstorm track, and ASUS UGen Intel Arc Pro B70 32GB for model integration.

Integration strategy

Validate the inference backend, model format and operator support before selecting an acceleration path.

Performance evaluation

Measure processing time, memory use and stage failures on a fixed lecture clip. Record any CPU fallback separately.

Hardware integration and performance evaluation are planned development stages.

Validation design

Question	Method	Measures
Are edits preserved?	Import, edit, export, reload	Field and timing consistency
Do captions improve?	A/B/C with human references	CER, repaired errors, new errors
Are explanations located?	Human explanation intervals	Start error, hits and omissions
Is retrieval faster?	Balanced player comparison	Time, success and timeouts
Is deployment feasible?	Fixed models and clip	Runtime, memory, fallback

Planned pilot: a fixed Chinese lecture and 6–10 participants with 4–6 search tasks. No user-study results yet.

Application value and adoption

Initial adoption

Start with one course team. Students test retrieval tasks. Teachers and teaching assistants measure review effort.

Commercial hypotheses

Evaluate course-team subscriptions and institutional deployment after the pilot. Pricing and willingness to pay remain untested.

Cost and value per hour of lecture

Measure processing time, equipment use, storage and human correction minutes alongside observed usage.

There are no claimed customers, revenue forecasts or measured time savings.

Development roadmap

Stage	Outcome
Transcript workflow	Speech recognition integration and source-preserving review
Concept indexing	Evidence-linked explanation intervals and retrieval evaluation
Visual context	Board frames and terminology candidates for review
Hardware integration	Validated inference backend and performance measurements
Course pilot	Retrieval tasks, user feedback and review-effort measurements

Each stage will be evaluated against source alignment, reproducibility and practical review tasks.

Reliability and research risks

Incorrect visual context

Keep the original caption and frame. Allow abstention. Evaluate formulas separately from terminology.

Wrong explanation start

Distinguish a first mention from actual teaching. Preserve evidence and allow manual boundary changes.

Lost or invalid data

Test export and reload. Reject invalid times. Mark generation failures instead of hiding them as empty results.

Performance claims

Separate sample workflows, model tests and user studies. Test the selected hardware before claiming acceleration.

Project resources

Application experience

Transcript search, synchronized captions and direct navigation to the corresponding lecture timestamp.

Demonstration

An English-captioned walkthrough illustrates search, bilingual queries and source-aligned video navigation.

Source and documentation

https://github.com/Windleaf1A/lecture-replay-assistant
README, demonstration files and verification records are included in the repository.

The walkthrough presents captured application states. The illustrated sample uses manually authored captions.

References and project documentation

Project implementation

Lecture Replay Assistant repository: software architecture, source code and verification record.
github.com/Windleaf1A/lecture-replay-assistant

Demonstration material

Original illustrated lesson and bilingual SRT captions. Asset provenance and the generation source are provided with the application.

Competition reference

ASUS UGen AI League official event page:
contest.bhuntr.com/tw/39jg9vimiynrhlksze/home/